

Trustworthy AI

AI that is lawful, ethical and technically robust across its whole lifecycle

GLOSSARY 06 - SHEET 8 / 8

PROFESSIONAL DEFINITION

Trustworthy AI is artificial intelligence that satisfies three components throughout its lifecycle: it is lawful, complying with applicable law; ethical, respecting ethical principles and values; and robust, both technically and socially, since even well-intentioned systems can cause unintentional harm. The three components are necessary and not individually sufficient, and the framework is operationalised through seven requirements with an accompanying assessment list (HLEG AI, 2019).

WHO DEFINED IT ORIGINALLY

The framework was set out by the European Commission's High-Level Expert Group on Artificial Intelligence in the Ethics Guidelines for Trustworthy AI (April 2019), with the Assessment List for Trustworthy AI following in July 2020. It provided the conceptual foundation for the White Paper on AI (2020) and, subsequently, for the AI Act.

WHY THIS DEFINITION WAS DEVELOPED

Ethics guidance was proliferating without any bridge to law or to engineering practice, so ethical commitments and technical assurance ran on separate tracks. Combining lawfulness, ethics and robustness in a single definition made the point that legal compliance alone is insufficient and that ethical intent without technical robustness produces harm regardless.

FIGURE 1 - THREE NECESSARY COMPONENTS

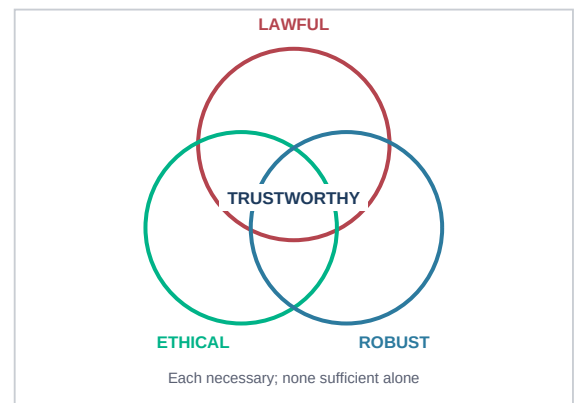


Figure 1. A system may be lawful and ethical and still cause harm through technical fragility; it may be robust and lawful and still be unacceptable. Only the intersection satisfies the definition, and the framework requires that conflicts between the three be recorded rather than resolved silently (after HLEG AI, 2019).

HOW IT IS USED

Assess the system against the seven requirements using the assessment list at design, at deployment and at review; record the trade-offs made between requirements and treat the record as the evidence of due diligence.

WHEN IT IS USED

At AI system design, before market placement, at periodic review of a live system, in vendor assessment, and in preparation for conformity assessment under the AI Act.

BY WHOM IT IS USED

AI ethics and governance boards, product and engineering leads, procurement functions assessing vendors, and executives accountable for the deployment decision.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor instruments were principle declarations, which stated commitments without any operational test, and technical assurance standards, which tested robustness without reference to ethical or legal acceptability. The trustworthy AI framework joins the three and supplies an assessment list, converting a statement of values into an auditable process. Its acknowledged weakness is that it does not resolve conflicts between the requirements; it requires only that trade-offs be made explicitly and recorded.

RELATED DEFINITIONS IN THIS COURSE

AI Ethics - Unit 6

EU AI Act - Unit 6

AI Governance - Unit 6

Explainability - Unit 6

Human-in-the-Loop - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

High-Level Expert Group on Artificial Intelligence (2019) Ethics Guidelines for Trustworthy AI. Brussels: European Commission.

High-Level Expert Group on Artificial Intelligence (2020) The Assessment List for Trustworthy Artificial Intelligence (ALTAI). Brussels: European Commission.

European Commission (2020) White Paper on Artificial Intelligence: A European Approach to Excellence and Trust, COM(2020) 65 final. Brussels: European Commission.